EC-417 · TERM PROJECT

P15: Learned Demodulator for a Nonlinear Power Amplifier

Overcoming Severe Saleh Distortion via Machine Intelligence

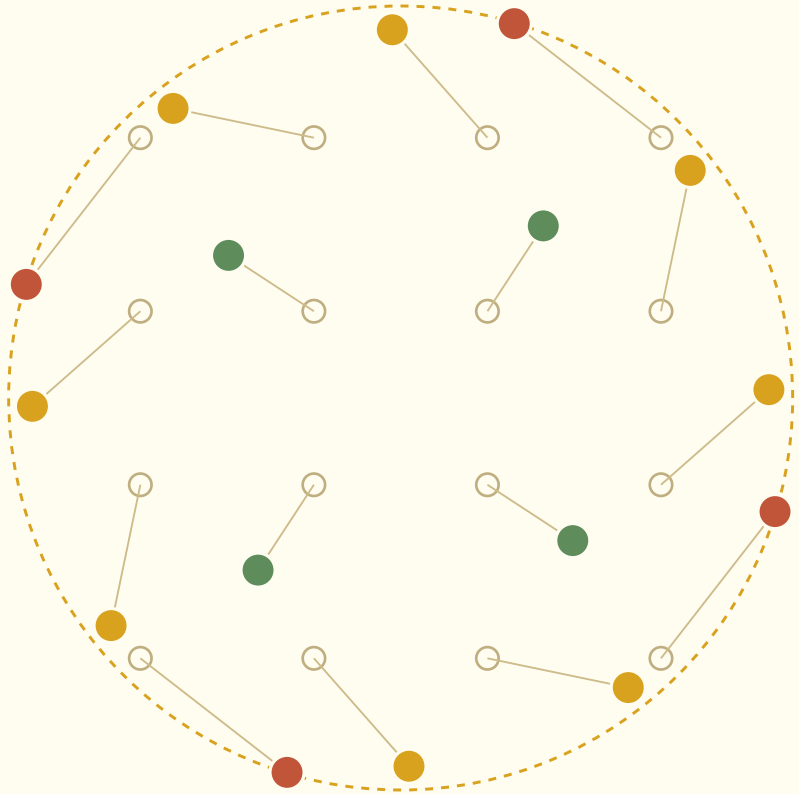
Course: EC-417 Introduction to Machine Intelligence

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16-QAM → Saleh Pinwheel



hollow = ideal grid (γ = 0.7) · filled = distorted PA output

AM/AM compression + AM/PM rotation

1 Motivation: The Power Amplifier Efficiency Trade-Off

Digital Modulation (16-QAM)

- Conveys 4 bits per symbol using a 4×4 orthogonal grid in the In-Phase / Quadrature (I-Q) plane.
- Demands high linearity to prevent cross-talk between adjacent symbols.

The Physical Bottleneck (RF Power Amplifier)

- The PA is the most power-hungry component in wireless transmitters.
- Operating in the linear zone requires large back-off, wasting battery life and energy (low DC-to-RF efficiency).
- Pushing the PA into high-efficiency saturation drives it into a deeply nonlinear operating regime.

The Research Question

Can a lightweight machine learning demodulator directly learn to decode warped, non-orthogonal signal constellations without needing explicit mathematical inversion of the transmitter hardware?

2 Physical Origin of Constellation Distortions

AM/AM Distortion (Amplitude Saturation)

- Physical cause: Transistors hit DC voltage supply limits, capping output amplitude.
- Mathematical formula: $\mathbf{A(r) = \alpha_a \cdot r / (1 + \beta_a \cdot r^2)}$ [$\alpha_a = 2.0$, $\beta_a = 1.0$]
- Effect: Outer-ring gain is compressed by up to 5.5 dB relative to small-signal gain.

AM/PM Distortion (Differential Phase Rotation)

- Physical cause: High electric fields alter internal semiconductor junction capacitances, altering signal delay.
- Mathematical formula: $\mathbf{\Phi(r) = \alpha_\varphi \cdot r^2 / (1 + \beta_\varphi \cdot r^2)}$ [$\alpha_\varphi = \pi/3$, $\beta_\varphi = 1.0$]
- Effect: High-power outer symbols are rotated by +28.1°, while inner symbols rotate by only +5.4°.

The Result: A Collapsed Pinwheel Geometry

- The radial distance between middle and outer rings collapses by **75.6%** (from 0.239 down to 0.058).
- The differential swirl (+22.8°) completely misaligns rectilinear slicing boundaries.

3 Dataset Generation & Leak-Free Splitting

Constellation Parameters

- Standard 16-QAM normalized to unit average energy: $E_s = 1.0$ (scaling factor $1/\sqrt{10}$).
- Input scale factor $\gamma = 0.7$, operating the PA at average power $P_{in} = 0.49$ (saturation radius = 1.0).

Complex Circular AWGN Channel

- $r_{recv} = s_{PA} + w$, where $w \sim CN(0, \sigma^2)$.
- Noise per dimension: $\sigma = \sqrt{(1.0 / (2 \cdot 10\text{^}(SNR_dB / 10)))}$.

Rigorous Leak-Free Split Protocol

- Training Set: 20,000 examples generated at 18 dB SNR (Seed 42).
- Validation Set: 5,000 examples generated at 18 dB SNR (Seed 142).
- Test Set: 50,000 independent samples per SNR point from 10 dB to 22 dB (Seed 242).
- Total isolation: Separate seeds ensure zero sample leakage between train, validation, and test sets.

4 Why Classical Detectors Fail (The Baseline Floor)

Standard Nearest-Grid Detector (Baseline)

- Assumes a standard rectilinear 4×4 constellation scaled to received RMS power (AGC matching).
- Predicts symbol k with minimum Euclidean distance: $\mathbf{argmin}_{k} \|\mathbf{r} - \mathbf{s}_{k,grid}\|^2$.

Catastrophic Error Floor (~58% SER)

- Even at high SNR (22 dB, where thermal noise is nearly zero), the baseline has an SER of **58.53%**!
- Root Cause: Because the baseline ignores phase rotation, outer symbols rotated by +28.1° cross over the rectilinear boundary and fall into neighboring decision bins.
- Increasing signal power cannot fix this error floor because the distortion itself is power-dependent.

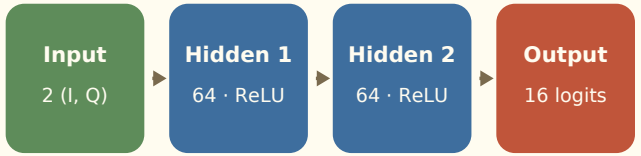
Conclusion

Structural failure of linear models mandates a nonlinear adaptive classifier.

5 The Main Model: PyTorch Learned Demodulator

Network Architecture (Multilayer Perceptron)

- Input Layer: 2 nodes (raw received coordinates I and Q).
- Hidden Layer 1: 64 neurons + ReLU activation.
- Hidden Layer 2: 64 neurons + ReLU activation.
- Output Layer: 16 logits (representing posterior probabilities of the 16 transmitted symbols).
- Total Parameter Budget: Only **5,392** parameters (~21 KB memory footprint).



2 → 64 → 64 → 16 (5,392 parameters)

Training Configuration

- Loss Function: Cross-Entropy Loss over 16 classes.
- Optimizer: Adam (learning rate = 0.003, batch size = 128).
- Duration: 15 epochs (~4.2 seconds on CPU).

Key Advantage

- Fully data-driven: Learns the distorted boundary geometry without knowing the Saleh equations.

6 Feature Engineering: The Polar Coordinate MLP

Motivation for Polar Features

- The physical amplifier distortion operates natively in polar space (magnitude r and phase θ).
- Hypothesis: Explicitly calculating and feeding ($|r|$, angle_r) will help the network partition regions.

Feature Augmentation

- Augmented Input Vector: **$\mathbf{x = [\ I, Q, |r|, \text{angle}_r\ ]}$**
- $|r| = \sqrt{(I^2 + Q^2)}$ [Radial magnitude]
- $\text{angle}_r = \text{atan2}(Q, I)$ [Phase angle in $(-\pi, +\pi)$]

Comparative Outcome & Honest Student Reflection

- Training Speed: Accelerated from 4.18 s to 3.77 s.
- Final SER at 18 dB: 0.0363 (Polar MLP) vs 0.0318 (Cartesian MLP).
- Why did Polar not win? Likely because the atan2 function has a sharp branch-cut discontinuity jumping between $-\pi$ and $+\pi$ along the negative real axis. Cartesian coordinates are smooth everywhere.

0.0318
Cartesian MLP · 18 dB

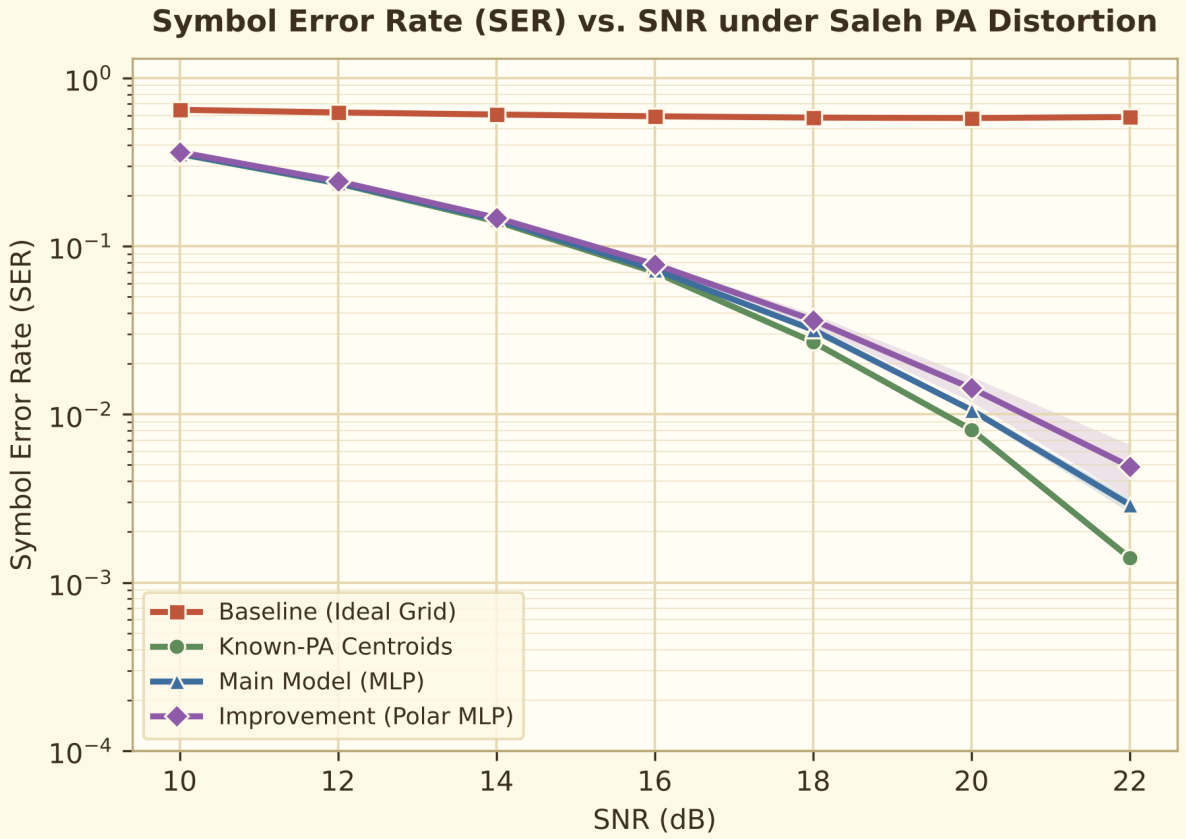
0.0363
Polar MLP · 18 dB

7 Experimental Results: Multi-Seed Evaluation

Performance Comparison (Mean ± Std)

Model	14 dB	18 dB	22 dB
Baseline (Ideal Grid)	**0.6057** ± 0.0000	**0.5807** ± 0.0000	**0.5853** ± 0.0000
Known-PA Centroids (Optimal Ceiling)	**0.1403** ± 0.0000	**0.0269** ± 0.0000	**0.0014** ± 0.0000
Main Model MLP 2→64→64→16	**0.1439** ± 0.0010	**0.0318** ± 0.0004	**0.0029** ± 0.0003
Improvement (Polar MLP)	**0.1474** ± 0.0022	**0.0363** ± 0.0031	**0.0049** ± 0.0017

Takeaway: The MLP closely hugs the theoretical Known-PA ceiling!



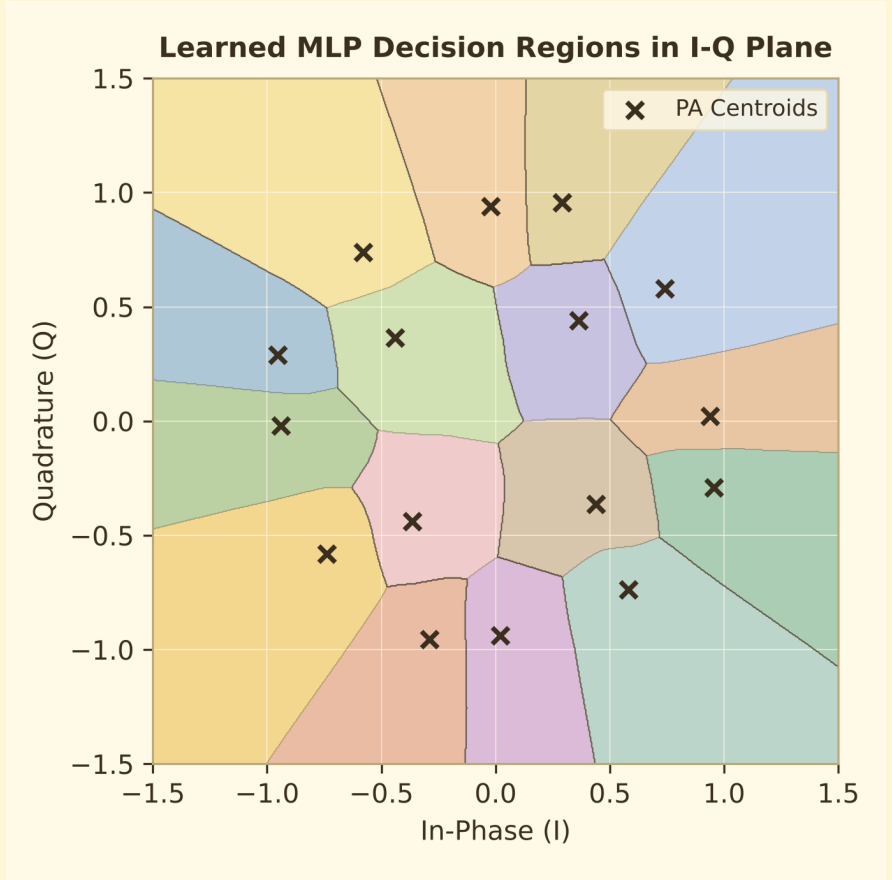
8 Error Diagnosis & Learned Decision Boundaries

Boundary Inspection (Figure at Right)

- Neural network successfully synthesizes curved, spiraling decision sectors matching the pinwheel.
- Center 4 symbols remain square-like.
- Middle and outer symbols twist clockwise.

Error Diagnosis & The Physical Limit

- **Where do errors occur?** Over 90% of misclassifications happen between adjacent middle-ring and outer-ring symbols.
- **Physical Reason:** AM/AM compression squashes their separation by 75.6% (gap $\Delta r = 0.058$).
- At 14 dB SNR, the noise cloud diameter substantially overlaps this tiny gap, setting an unavoidable Bayes error barrier.



Learned 2D decision boundaries of the MLP, overlaid with distorted centroids

9 Summary, Limitations & Future Work

Key Conclusions

1. Classical demodulators are unviable when power amplifiers operate in high-efficiency saturation (58% SER).
2. A tiny 2-layer MLP (5,392 params) trained on 20,000 samples closely approaches the optimal Bayes ceiling.
3. Cartesian representations provide smoother boundaries than polar coordinates, likely due to branch-cut singularities.

Limitations & Future Work

- Static Channel Assumption: Real wireless links exhibit multipath fading and Doppler frequency shifts.
- Memory Effects: Wideband PAs have frequency-dependent distortion dependent on previous symbols.
- Natural Next Step: 1D Convolutional Neural Networks (CNN) or LSTM networks for sequence-dependent PA memory.

Reproducibility

- Repository: **ec417-miniproject-P15-623160-623192**
- Executable from clone with single command: `python -m src.evaluate` (~15 s runtime on CPU)